

Howard Park

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PROFESSIONAL SUMMARY

Tools engineer with seven years in games, building the internal software other disciplines depend on. At Meta wrote a Unity sandbox tool that let sound engineers test audio profiles without an engineer in the loop, and optimized in-house asset creation and profiling tools for Horizon Worlds. At Ultraviolet owned the art pipeline between artists and the Unity client. Full-stack web experience across TypeScript, JavaScript and Python, with REST API design on AWS and a Supabase/PostgreSQL auth pipeline. Portfolio: hjp309.github.io

CORE COMPETENCIES

Internal Developer Tools

Full-Stack Web

REST API Design

TypeScript & JavaScript

Python

PostgreSQL & Supabase

Asset & Build Pipelines

Cross-discipline Collaboration

WORK EXPERIENCE

Meta December 2023 - December 2024

Technical Artist

- Wrote a Unity sandbox tool for sound engineers to test ambient and avatar sounds, optimizing noise profiles in high- and low-traffic settings.
- Improved performance for Horizon Worlds (100k+ users) on Unity by writing shaders and optimizing in-house asset creation and profiling tools.
- Performed asset conversions and visual fidelity tests between Horizon and Unity environments to improve asset creation pipelines.
- Worked cross-functionally with engineers, artists, and designers, scoping technical feasibility and requirements for new seasonal content.

FiveMics August 2022 - November 2023

Backend Engineer

- Developed secure REST APIs on API Gateway with Lambda functions to handle account registration, authentication, and user profile management.
- Produced clean documentation for accessing AWS projects and CLI utilization.
- Developed a full-stack onboarding and registration flow with UX artists and designers, supporting Desktop, Google and Apple sign-in via Cognito, DynamoDB and Unity.
- Employed the AWS CLI to deploy Unity builds on CloudFormation for scalable infrastructure updates.
- Organized user profiling with role-based access control policies and secret management using IAM and AWS Secrets Manager.

Ultraviolet April 2021 - June 2022

Unity Developer

- Owned the art pipeline, handling asset flow and scene staging between artists and the Unity client.
- Collaborated with iOS engineers to integrate Unity builds with Xcode and administer two-way communication using the Unity SDK.
- Collaborated with ML engineers to implement analytics APIs in the game to monitor player activity and retention.
- Led a small team of Unity developers, junior engineers, and contractors to build a metaverse prototype for the Octi social media app.
- Increased community engagement by 20%, reaching #1 in the social media category for two weeks.

Evara VR September 2020 - March 2021

Unity Developer (AR)

- Produced AR experiences using ARKit and Unity for venues including South Coast Plaza.

Konami Digital Entertainment

December 2019 - March 2020

Quality Assurance Tester

- Reported and tracked bugs with clear replication steps, and conducted console TRC compliance tests for several titles including *Yu-Gi-Oh! Legacy of the Duelist: Link Evolution* and *Skelattack*.

Tandon Online

June 2018 - August 2018

Unity Developer

- Created an online educational game where students collaboratively built the tallest tower possible, developed with school faculty to teach organizational behavior principles.
- Networked all server-client interactions using Unity HLAPI, including player-specific blocks, crane mechanics, and user interface.

PROJECTS

LoL Draft Simulator: Supabase, Data Management, Full-stack Unity

Secure registration and login pipeline built on the Supabase C# SDK, with TypeScript automating real-time data manipulation. Python scripts scrape and import millions of player and game metrics via the CargoQuery API, feeding a scikit-learn regression model that scores roster strength.

Technologies: Supabase, TypeScript, PostgreSQL, Python, Unity/C#

Firebot Auto Battler: Live-Service Systems, Twitch Integration

Twitch-integrated auto-battler running as a Firebot custom script. Persistent viewer accounts and weapon state across sessions, a currency system, and Twitch Prediction betting that resolves and pays out against battle results, with a real-time OBS overlay.

Technologies: Firebot, Twitch API (events, predictions), JavaScript, OBS overlay

WishSeekers: Solo Gameplay Programmer, Unity 6 / C#

Sole engineer on the full gameplay codebase of a 2.5D action roguelite in development. Built on a ScriptableObject data layer with all damage routed through one resolution pipeline (modifiers, crit, mitigation, absorb, afflictions) shared by 30+ abilities. Four NavMesh enemy archetypes and a multi-phase boss using animation-event hitboxes, telegraph windows and coroutine state machines. Game-feel stack spanning a melee combo state machine, i-frames, parry/counter and an input buffer with press-grace and release-edge windows, plus 9 custom URP/HLSL shaders, auto-registering object pools, Addressables scene streaming, custom editor tools and an Inspector-level balance guide for non-programmers.

Technologies: Unity 6/C#, URP/HLSL, ScriptableObjects, NavMesh, Addressables

Limb From Limb: Game Jam, Engineering Lead

Led engineering on a four-person team to ship a complete 2D shooter-platformer in two weeks, in a \$5K game jam with 343 entries. Scoped and delivered the full mechanic, combat and platforming systems under a hard deadline.

Technologies: Unity/C#

SKILLS

Web & Backend: TypeScript, JavaScript, Python, REST API design, Supabase, PostgreSQL, AWS (API Gateway, Lambda, DynamoDB, Cognito)

Tooling: Unity editor tooling, asset creation and profiling tools, asset conversion and fidelity testing, art and build pipelines, technical documentation

Languages: C#, TypeScript, JavaScript, Python, PostgreSQL, C++, Java, HLSL

Data: Pandas, NumPy, scikit-learn, Selenium, BeautifulSoup, CargoQuery API

EDUCATION

B.S. Integrated Digital Media, Minor in Game Engineering

2015 - 2019

New York University, Tandon School of Engineering

Dean's List 2016-2017. Relevant coursework: Game Programming, Data Structures & Algorithms, Object Oriented Programming, Artificial Intelligence, Ideation & Prototyping.